PRESSURE SENSOR SYSTEM

Using Load Cell, HX711 Amplifier & ESP32 Microcontroller

Project Report

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Table of Contents

1. Introduction 2
2. Objectives 3
3. Components Used 3
4. Working Principle 4
5. System Architecture 5
6. Calibration Process 5
7. Challenges Faced 6
8. Project Setup & Hardware 6
9. Results & Discussion 7
10. Conclusion 7

1. Introduction

This project was developed as part of the Sensors and Signal Conditioning course during the 3rd semester. The objective was to design and implement a real-time pressure detection system using a load cell as the primary sensing element. The system measures applied force, processes it through an amplifier and microcontroller, and displays the result on an LCD screen.

Pressure measurement is fundamental to many engineering and industrial applications, including weighing systems, structural health monitoring, and biomedical instrumentation. This project demonstrates a practical, low-cost implementation of such a system using readily available components mounted on a cardboard platform.

2. Objectives

- To detect applied pressure/force using a load cell sensor.
- To amplify the small analog signal from the load cell using the HX711 module.
- To convert the raw digital data to meaningful force/pressure values using the ESP32.
- To display the calculated pressure value in real-time on a 16×2 LCD.
- To calibrate the system against known weights for accuracy and repeatability.
- To build a compact, structured prototype using a cardboard platform.

3. Components Used

Component	Quantity	Purpose
Load Cell Sensor	1	Primary force/pressure sensing element
HX711 Amplifier Module	1	Amplifies and digitizes the strain gauge signal
ESP32 Microcontroller	1	Processes raw data; converts to pressure value
16×2 LCD Display	1	Displays real-time pressure readings
Cardboard Platform	1	Structural base/housing for components
Connecting Wires	Several	Electrical interconnections
Wooden/Cardboard Disc	1	Force application platform on load cell

4. Working Principle

The system operates on the principle of resistive strain measurement. The load cell contains a strain gauge — a thin metallic foil pattern bonded to a flexible backing. When a force is applied to the load cell, it deflects slightly, causing the strain gauge to deform. This deformation changes the electrical resistance of the gauge, producing a small voltage output proportional to the applied force via the Wheatstone bridge circuit inside the load cell.

The signal chain is as follows:

- Force is applied to the cardboard disc on top of the load cell.
- The load cell deflects, and the internal strain gauge generates a small differential voltage (typically in the microvolt to millivolt range).
- The HX711 amplifier module receives this signal, amplifies it with a fixed gain (64× or 128×), and converts it to a 24-bit digital value via an internal Analog-to-Digital Converter (ADC).
- The ESP32 reads this raw digital value from the HX711 via its clock and data pins.
- The ESP32 firmware applies a calibration factor to convert the raw count to a meaningful force (in grams or Newtons) using a mathematical formula.
- The computed pressure/force value is displayed on the 16×2 LCD screen in real-time.

5. System Architecture

The block diagram below illustrates the signal flow from sensor to display:

6. Calibration Process

Calibration was the most critical and challenging step in this project. The raw output from the HX711 is a unitless digital count that must be mapped to a known physical quantity. The calibration procedure followed was:

- Step 1 – Zero Offset: With no load applied, record the raw ADC count (tare value). This offsets the baseline reading.
- Step 2 – Known Weight Placement: Place a known reference weight (e.g., 100 g or 500 g) on the load cell platform.
- Step 3 – Record Raw Count: Note the ADC raw count with the known weight applied.
- Step 4 – Compute Calibration Factor: Divide the difference in raw counts by the known weight to get the scale factor (counts per gram).
- Step 5 – Code Integration: Program this factor into the ESP32 firmware so every future reading is automatically converted to grams or Newtons.

- Step 6 – Verify Repeatability: Apply the known weight multiple times and check for consistent readings.

Formula used in firmware:

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Force (g) = (Raw Count − Tare Value) ÷ Calibration Factor
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7. Challenges Faced

- Raw Data Mapping: The most significant challenge was accurately mapping the raw ADC output of the HX711 to a real weight value. Minor errors in the calibration factor led to incorrect readings, requiring repeated iterations.
- Signal Noise: The load cell signal is very sensitive to vibration and electrical interference. Keeping wires short and away from noise sources helped mitigate this.
- Repeatability Issues: Initial tests showed slight variation on repeated measurements. This was resolved by averaging multiple readings in the ESP32 firmware.
- Structural Stability: The cardboard platform had to be rigid enough to ensure consistent loading and not flex under weight, which could introduce measurement error.
- LCD Integration: Displaying the data clearly with appropriate labels and units (grams / Newtons) required careful LCD library configuration.

8. Project Setup & Hardware

The following images show the actual hardware prototype built for this project:

Figure 1: Load Cell sensor mounted on cardboard platform with force application disc

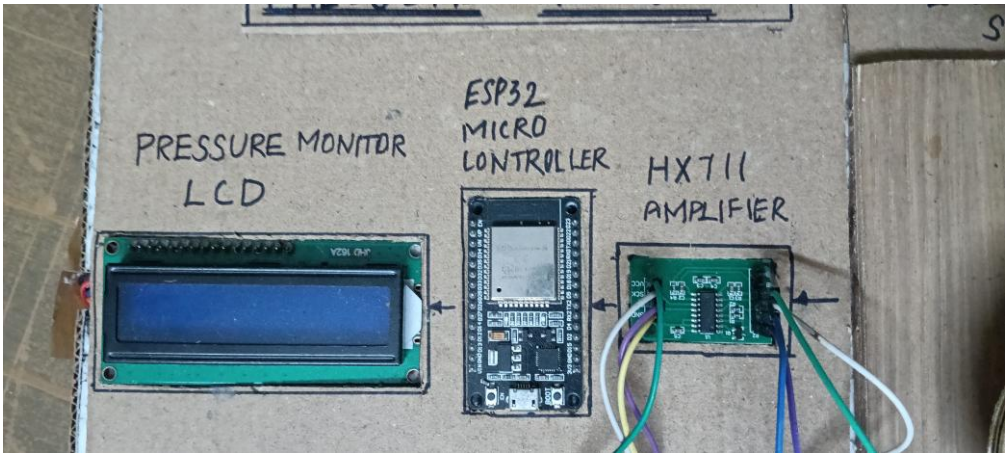

Figure 2: Electronics panel – Pressure Monitor LCD, ESP32 Microcontroller, and HX711 Amplifier

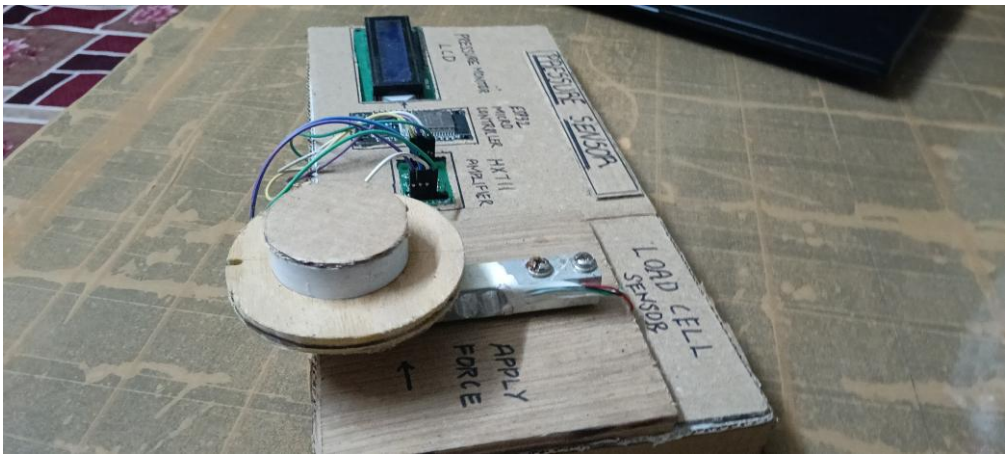

Figure 3: Side view showing the full assembly – load cell, disc, and electronics panel

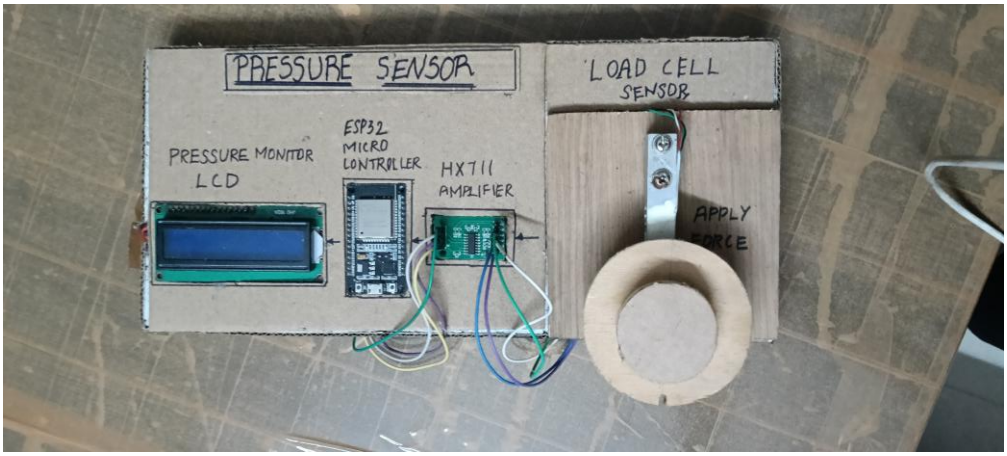

Figure 4: Top view of the complete Pressure Sensor prototype

9. Results & Discussion

After calibration, the system successfully detected applied force and displayed corresponding pressure values on the LCD in real-time. The key observations were:

- The system responded immediately when force was applied to the disc.
- After calibration with a known reference weight, readings were accurate within an acceptable tolerance range.
- Averaging multiple HX711 samples in the ESP32 code significantly improved repeatability.
- The LCD clearly displayed the measured force value, confirming end-to-end functionality of the signal chain.
- The prototype demonstrated that a low-cost load cell + HX711 + ESP32 combination can be used to build functional pressure measurement systems.

10. Conclusion

This project successfully demonstrated the design and implementation of a pressure detection system using a load cell as the sensing element, the HX711 as the signal conditioning amplifier, and the ESP32 as the processing and display controller. The system was built on a cardboard platform, making it cost-effective and easy to assemble.

The project deepened our understanding of sensor characteristics, signal conditioning principles, analog-to-digital conversion, and calibration techniques — all of which are core concepts of the Sensors and Signal Conditioning course. The major learning from this project was the importance of accurate calibration in converting raw sensor data into meaningful, repeatable measurements.

> *Key Takeaway: A well-calibrated load cell system using the HX711 and ESP32 can provide accurate, real-time pressure measurements, suitable for educational demonstrations and low-cost weighing applications.*